Research Methodology & Intellectual Property Rights

Course Code: 22URMK506
Course Hours: 26

Reference Books:

1. C.R. Kothari, Gaurav Garg, Research Methodology: Methods and Techniques, New Age International, 4th Edition, 2018.
2. Ranjit Kumar, Research Methodology a step-by-step guide for beginners, SAGE Publications, 3rd Edition, 2011.
3. Fink A, Conducting Research Literature Reviews: From the Internet to Paper, Sage Publications, 2009.
4. N. K. Acharya, Text book on Intellectual Property Rights, 4th Edition, Asia Law House, Hyderabad.

UNIT - I

Research Methodology

- Composed of two syllables "Re" and "Search".
- "Re" ='Again or over again or a new' and
- "Search" = is the latter meaning 'to examine closely and carefully' or 'to test and try'.

Research can also be defined as

- 1. Search for knowledge
- 2. Systematic and scientific search for getting relevant answers on any taken up specific topic.
- 3. Scientific enquiry into a subject.
- 4. Research is a movement from the unknown to the known.
- 5. It is the voyage of discovery

Research comprises of

- Defining and redefining problems.
- Formulating hypothesis (basic idea)
- Collecting data
- Organizing data
- Evaluating data
- Making decisions
- Suggesting solutions
- Reaching conclusions
- Finally, carefully testing the conclusions

1.1 MEANING OF RESEARCH

- ✓ Research in common parlance refers to a **search for knowledge**.
- ✓ One can also define research as a scientific and systematic search for pertinent information on a specific topic.
- ✓ In fact, research is an art of **scientific investigation.**
- ✓ Research is an academic activity and as such the term should be used in a technical sense.

The Advanced Learner's Dictionary of Current English lays down the meaning of research as,

1. **"A careful investigation or inquiry specially through search for new facts in any branch of knowledge."**

2. Redman and Mory define research as a **"systematized effort to gain new knowledge."**

3. M. Stephenson in the Encyclopaedia of Social Sciences define research as **"the manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in the practice of an art."**

Research is, thus, an original contribution to the existing stock of knowledge making for its advancement.

The term 'research' refers to the systematic method consisting of the problem, formulating a hypothesis, collecting the facts or data, analyzing the facts and reaching certain conclusions either in the form of solutions(s) towards the concerned problem or in certain generalizations for some theoretical formulation.

1.2 OBJECTIVES OF RESEARCH:

The purpose of research is to discover answers to questions through the application of scientific procedures.

The main aim of research is to find out the **truth** which is hidden and which has not been **discovered as yet**. Though each research study has its own specific purpose.

we may think of research objectives as falling into a number of following broad groupings:

1. **To gain familiarity with a phenomenon or to achieve new insights into it** (studies with this object in view are termed as exploratory or formulative research studies);

2. **To portray accurately the characteristics of a particular individual, situation or a group** (studies with this object in view are known as descriptive research studies);

3. **To determine the frequency with which something occurs or with which it is associated with something else** (studies with this object in view are known as diagnostic research studies);

4. **To test a hypothesis of a causal relationship between variables** (such studies are known as hypothesis-testing research studies).

Recap

What is Research?
Objectives of Research.

1.3 MOTIVATION IN RESEARCH

What makes people to undertake research? This is a question of fundamental importance. The possible motives for doing research may be either one or more of the following:

1. Desire to get a research degree along with its consequential benefits;
2. Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiates research;

3. Desire to get intellectual joy of doing some creative work;
4. Desire to be of service to society;
5. Desire to get respectability.

However, this is not an exhaustive list of factors motivating people to undertake research studies. Many more factors such as directives of government, employment conditions, curiosity about new things, desire to understand causal relationships, social thinking and awakening, and the like may as well motivate (or at times compel) people to perform research operations.

1.4 TYPES OF RESEARCH

1. Descriptive vs. Analytical
2. Applied vs. Fundamental
3. Quantitative vs. Qualitative
4. Conceptual vs. Empirical
5. Some Other Types of Research

1. Descriptive vs. Analytical:

- **Descriptive Research:**

  Purpose: To describe the characteristics or functions of a specific phenomenon.

  Approach: Observes and documents **what is happening** without making predictions or establishing causal relationships.

  Example: A survey to determine the preferences of smartphone users.

- **Analytical Research:**

  Purpose: To understand the reasons or causes behind a phenomenon, often involving hypothesis testing.(Hypothesis testing is a systematic procedure for deciding whether the results of a research study support a particular theory which applies to a population.)

  Approach: Analyses data and relationships between variables to explain **"why" and "how"** something happens.

  Example: A study that examines the relationship between diet and heart disease.

2. Applied vs. Fundamental:

- **Applied Research:**

Purpose: To solve specific, practical problems and find solutions that can be applied immediately.
Approach: Often involves real-world applications, focusing on practical outcomes.
Example: Developing a new software tool to improve online learning.

- **Fundamental (Basic or Pure) Research:**

Purpose: To expand knowledge by understanding fundamental principles, without immediate application.
Approach: Theoretical, aiming to increase understanding of basic concepts.
Example: Researching the properties of a newly discovered chemical element.

3. Quantitative vs. Qualitative:

- **Quantitative Research:**

  Purpose: To quantify data and generalize results from a sample to the population of interest.

  Approach: Uses structured tools like surveys, experiments, and statistical analysis to gather numerical data.

  Example: Measuring the effectiveness of a new drug through clinical trials.

- **Qualitative Research:**

  Purpose: To understand the underlying reasons, opinions, and motivations behind a phenomenon.

  Approach: Uses unstructured or semi-structured techniques like interviews, focus groups, and content analysis to gather non-numerical data.

  Example: Exploring patient experiences with chronic illness through in-depth interviews.

4. Conceptual vs. Empirical:

- **Conceptual Research:**
  Purpose: To develop new theories or concepts without the use of empirical data.
  Approach: Based on abstract ideas, literature review, and logical reasoning.
  Example: Formulating a new theory on organizational behavior based on existing literature.

- **Empirical Research:**
  Purpose: To test hypotheses or theories using observed and measured phenomena.
  Approach: Relies on data collection through observation, experimentation, or surveys.
  Example: Conducting experiments to test the impact of sleep deprivation on cognitive performance.

5. Some Other Types of Research:

- **Exploratory Research:**

  Purpose: To explore a problem that has not been clearly defined or understood.

  Approach: Often involves gathering preliminary data that helps define the problem and suggests hypotheses.

  Example: A pilot study investigating the feasibility of a new teaching method.

- **Cross-sectional Research:**

  Purpose: To analyze data from a population at one specific point in time.

  Approach: Involves observing a defined population to gather data on various variables at a single moment.

  Example: A survey conducted to understand the current opinions on climate change.

- **Longitudinal Research:**

  Purpose: To study the same subjects over a period of time to observe changes and developments.

  Approach: Data is collected repeatedly from the same subjects at different times.

  Example: A study tracking the health outcomes of participants over a decade.

- **Experimental Research:**

  Purpose: To establish cause-and-effect relationships by manipulating one or more variables.

  Approach: Involves controlled testing with an experimental group and a control group.

  Example: Testing the effectiveness of a new drug in a controlled environment.

- **Action Research:**

  Purpose: To solve an immediate problem while simultaneously contributing to theory and practice.

  Approach: Involves cycles of planning, action, observation, and reflection.

  Example: Teachers adapting teaching methods based on classroom observations and student feedback.

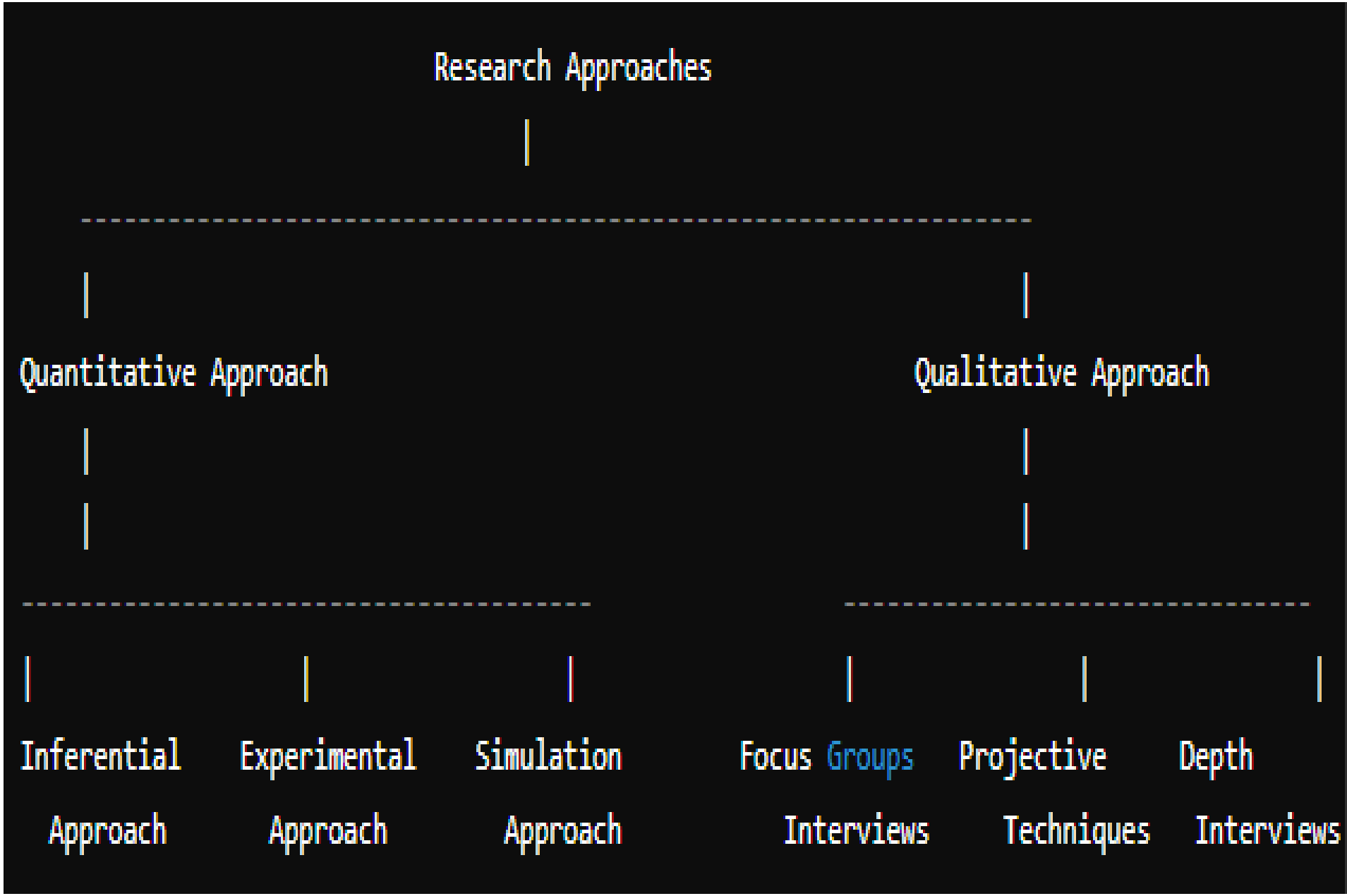
- **Case Study Research:**

  Purpose: To conduct an in-depth analysis of a single case or a small number of cases.

  Approach: Focuses on detailed contextual analysis.

  Example: A study analysing the business strategies of a successful startup.

1.5 Research Approaches



The classification of research approaches can be broadly divided into two main categories: **Quantitative Approach** and **Qualitative Approach**.

1. Quantitative Approach

The quantitative approach involves generating data in numerical form, which can then be **analyzed statistically and rigorously**. This approach is characterized by **its structured and formal methodology**. It is sub-classified into the following:

- ✓ **Inferential Approach**:

  Focuses on forming a database from which to infer characteristics or relationships within a population.

  Typically involves survey research, where a sample is studied to determine its characteristics, and these findings are then generalized to the broader population.

- ✓ **Experimental Approach:**

Involves controlling the research environment more strictly.

Some variables are manipulated to observe their effects on other variables.

✓ **Simulation Approach:**

Involves creating an artificial environment to generate relevant data.
Allows observation of a system's dynamic behavior under controlled conditions.
Often used in business and social sciences to model and understand future conditions or complex processes.

2. Qualitative Approach

The qualitative approach is concerned with the subjective assessment of attitudes, opinions, and behaviors. This approach is less structured and relies more on the researcher's insights and impressions. The results are generally in non-numerical form and are not subjected to rigorous quantitative analysis. It includes the following techniques:

- ✓ **Focus Group Interviews:** Small groups of people are interviewed to gather diverse opinions and insights on a particular topic.
- ✓ **Projective Techniques:** Projective techniques are indirect ways to understand people's hidden feelings and attitudes. Instead of asking them directly, these methods present them with unclear or vague situations, pictures, or words. People then "project" their inner thoughts and emotions onto these unclear things, which helps reveal what they might not say outright.
- ✓ **Depth Interviews:** Intensive, one-to-one interviews that explore deep-seated motivations, feelings, and behaviors.

1.6 SIGNIFICANCE OF RESEARCH

1. Research provides the basis for nearly all government policies in our economic system.
2. Research has its special significance in solving various operational and planning problems of business and industry.
3. Research is equally important for social scientists in studying social relationships and in seeking answers to various social problems.

In addition to what has been stated above, the significance of research can also be understood keeping in view the following points:

- ✓ To those students who are to write a master's or Ph.D. thesis, research may mean a self-seeking (selfish, self-interested, calculative) or a way to attain a high position in the social structure;
- ✓ To professionals in research methodology, research may mean a source of livelihood;
- ✓ To philosophers and thinkers, research may mean the outlet for **new** ideas and insights;
- ✓ To literary men and women, research may mean the development of new styles and creative work;
- ✓ To analysts and intellectuals, research may mean the generalizations of new theories.

Thus, research is the fountain of knowledge for the sake of knowledge and an important source of providing guidelines for solving different business, governmental and social problems. It is a sort of formal training which enables one to understand the new developments in one's field in a better way.

1.7 RESEARCH METHODS VERSUS METHODOLOGY

Aspect	Research Methods	Methodology
Definition	Specific techniques or procedures used to gather and analyze data in a research study.	The overall approach, reasoning, or strategy guiding the choice and use of methods in a research study.
Focus	Focuses on the **how**: how to collect and analyze data (e.g. Surveys, experiments, interviews).	Focuses on the **why**: the underlying logic or framework of the research process (e.g., why certain methods are chosen).
Scope	**Narrower:** concerned with the tools and techniques used to carry out the research.	**Broader:** concerned with the overall design, including the choice of methods and justification.

Purpose	To provide specific tools for data collection and analysis.	To guide the research process and provide a reason, thinking, logic, for selecting specific methods.
Examples	Surveys, interviews, observations, experiments, statistical analysis.	Qualitative, quantitative, or mixed methods approaches; grounded theory, phenomenology.
Level of Abstraction	Concrete and practical, focusing on action steps and procedures.	More abstract, dealing with philosophical assumptions and research strategy.
Associated with	Data collection and data analysis techniques.	Research design, philosophy, and framework of inquiry.

Application	Used in specific stages of research to collect and process data.	Applied throughout the research design to determine which methods will be used and how they align with research objectives.
Key Questions Addressed	What techniques will be used to collect and analyze the data?	Why are these methods chosen, and how do they align with the research question or problem?
Examples of Key Terms	Sampling, data collection, coding, statistical analysis.	Positivism, interpretivism, epistemology, research paradigms.

In essence, research methods are the specific procedures used to gather and analyze data, while methodology refers to the broader philosophical approach that justifies and guides the selection and use of those methods.

1.8 DEFINING THE RESEARCH PROBLEM:

In research process, the first and foremost step happens to be that of selecting and properly defining a research problem.

A researcher must find the problem and formulate it so that it becomes susceptible to research. Like a medical doctor, a researcher must examine all the symptoms (presented to him or observed by him) concerning a problem before he can diagnose correctly.

To define a problem correctly, a researcher must know: what a problem is?

❑ WHAT IS A RESEARCH PROBLEM?

A research problem, in general, refers to some difficulty which a researcher experiences in the context of either a theoretical or practical situation and wants to obtain a solution for the same.

We can, thus, state the components of a research problem as under:

1. There must be an individual or a group which has some difficulty or the problem.
2. There must be some objective(s) to be attained at. If one wants nothing, one cannot have a problem.
3. There must be alternative means (or the courses of action) for obtaining the objective(s) one wishes to attain. This means that there must be *at least two means* available to a researcher for if he has no choice of means, he cannot have a problem.
4. There must remain some doubt in the mind of a researcher with regard to the selection of alternatives. This means that research must answer the question concerning the relative efficiency of the possible alternatives.
5. There must be some environment(s) to which the difficulty pertains.

Thus, a research problem is one which requires a researcher to find out the best solution for the given problem,

1.9 SELECTING THE PROBLEM

The research problem undertaken for study must be carefully selected. The task is a difficult one, although it may not appear to be so. Help may be taken from a research guide in this connection.

Nevertheless, every researcher must find out his own salvation for research problems cannot be borrowed.

Thus, a research guide can at the most only help a researcher choose a subject. However, the following points may be observed by a researcher in selecting a research problem or a subject for research:

1. Subject which is overdone should not be normally chosen, for it will be a difficult task to throw any new light in such a case.
2. Controversial subject should not become the choice of an average researcher.
3. Too narrow or too vague problems should be avoided.
4. The subject selected for research should be familiar and feasible so that the related research material or sources of research are within one's reach.

5. The importance of the subject, the qualifications and the training of a researcher, the costs involved, the time factor are few other criteria that must also be considered in selecting a problem. In other words, before the final selection of a problem is done, a researcher must ask himself the following questions:

(a) Whether he is well equipped in terms of his background to carry out the research?

(b) Whether the study falls within the budget he can afford?

(c) Whether the necessary cooperation can be obtained from those who must participate in research as subjects?

6. The selection of a problem must be preceded by a preliminary study.

If the subject for research is selected properly by observing the above mentioned points, the research will not be a boring The subject or the problem selected must involve the researcher and must have an upper most place in his mind so that he may undertake all pains needed for the study.

1.10 NECESSITY OF DEFINING THE PROBLEM:

1. Clarity and Focus

A well-defined problem provides clarity on what exactly needs to be solved. Without this, efforts can become scattered or misdirected. It helps focus on the root cause rather than just symptoms.

2. Effective Use of Resources

When the problem is clearly defined, time, money, and human resources can be allocated effectively. Misidentifying the problem leads to wasted efforts on irrelevant areas.

3. Objective Decision-Making

A precise problem definition ensures that decisions and solutions are based on the actual issue rather than assumptions or misunderstandings. It reduces the risk of biases affecting the solution.

4. Enhanced Communication
Clearly defining a problem makes it easier to communicate with team members, stakeholders, or clients. It aligns everyone's understanding of the issue and the goals.

5. Measurable Goals and Solutions
Defining the problem helps establish clear criteria for success. It sets the foundation for measurable outcomes and allows you to evaluate the effectiveness of the solutions implemented.

6. Prevents Scope Creep
In projects, a clearly defined problem prevents scope creep, where the objectives of the project slowly expand or shift without proper control. This ensures the project remains focused and manageable.

7. Foundation for Analysis

Once the problem is defined, it's easier to break it down and analyze it. You can apply structured problem-solving methods, such as root cause analysis, to identify the underlying issues and develop targeted solutions. In short, defining the problem is essential for solving it effectively, efficiently, and in a focused manner.

1.11 TECHNIQUE INVOLVED IN DEFINING A PROBLEM

Defining a research problem properly and clearly is a crucial part of a research study and must in no case be accomplished hurriedly. However, in practice this a frequently overlooked which causes a lot of problems later on. Hence, the research problem should be defined in a systematic manner, giving due weightage to all relating points.

The technique for the purpose involves the undertaking of the following steps generally one after the other:

1. **Statement of the problem in a general way;**
2. **Understanding the nature of the problem;**
3. **Surveying the available literature**
4. **Developing the ideas through discussions; and**
5. **Rephrasing the research problem into a working proposition.**

1. Statement of the problem in a general way:
This involves defining the broad issue or question that needs investigation. It helps in setting the context for the research by identifying what needs to be solved or explored.

2. Understanding the nature of the problem: After stating the problem, it's crucial to analyze it thoroughly. This step involves understanding the problem's scope, complexity, and factors involved to ensure clarity.

3. Surveying the available literature:

A literature review is conducted to explore existing research related to the problem. This helps in identifying gaps, methodologies used, and insights from previous studies.

4. Developing the ideas through discussions:

Engaging in discussions with peers, experts, or mentors can refine your ideas. It helps to brainstorm various approaches and perspectives to better shape the research problem.

5. Rephrasing the research problem into a working proposition:

After understanding the problem and reviewing literature, the research problem is redefined into a more specific, clear, and researchable statement. This working proposition guides the direction of the research.